

Week 11 — Computer Vision

Deep Learning for Visual Understanding

Resource Guide & Learning Toolkit

Deep Learning · Fusemachines AI Fellowship
Facilitator: Susan Ghimire | Dataset: CIFAR-10 (torchvision) + COCO-pretrained models

How to use this guide: Organised in five sections mirroring the module phases. **Step 1** — Watch the priority videos. **Step 2** — Read the corresponding chapter or paper — free, online. **Step 3** — Work through the Kaggle tutorials for hands-on practice. **Step 4** — Use the AI prompts: paste each into Claude or ChatGPT, then apply the critique guidance to verify the answer. **Step 5** — Bring the discussion questions to your mid-week sync. All links are clickable. Resources marked ★ **Priority** are the minimum required. This guide covers the topic broadly — the specific notebook questions and submission requirements are in the Project Guide.

LEARNING PHASES

Phase	Topics
1 <i>CNN Foundations & Image Classification</i>	• Convolution, pooling, and receptive fields as learned feature hierarchies • Transfer learning: freezing a pretrained backbone vs fine-tuning end-to-end • ImageNet-pretrained backbones (ResNet-50) as general-purpose feature extractors • GradCAM: gradient-weighted class activation maps for explainability • Connecting hand-crafted kernels (Sobel, blur) to learned convolutional filters
2 <i>Object Detection</i>	• Intersection over Union (IoU) as the core box-overlap metric • Anchor boxes and region proposal networks • Two-stage detectors (Faster R-CNN) vs single-stage detectors (YOLO) • Non-Maximum Suppression for de-duplicating overlapping detections • Using COCO-pretrained detection models directly for inference
3 <i>Image Segmentation</i>	• Semantic segmentation vs instance segmentation • Encoder–decoder architectures with skip connections (U-Net, DeepLabv3+) • Pixel-level evaluation: pixel accuracy, mean Intersection over Union • Classical morphological pipelines vs learned segmentation — trade-offs
4 <i>Generative Models — Variational Autoencoders</i>	• Latent variable models and the reparameterisation trick • ELBO: balancing reconstruction loss against KL divergence • Latent space structure and interpolation between samples • Synthetic data generation as an augmentation strategy for low-data classes
5 <i>Vision Transformers & Deployment</i>	• Patch embedding: treating an image as a sequence of tokens • Self-attention vs convolutional inductive bias — when each wins • CLIP: contrastive vision-language pretraining and zero-shot classification • ONNX export and runtime-agnostic inference for production deployment

RESOURCES

★ Priority Videos — Watch Before the Session

► Andrej Karpathy — The Spelled-Out Intro to Neural Networks and Backpropagation

<https://www.youtube.com/watch?v=VMj-3S1tku0>

★ Priority · Phase 1. Builds a neural network and backpropagation from scratch in Python. Foundational for understanding what a CNN's learned weights actually represent. Long (~2.5 hr) — the first hour covers the essentials.

► But What Is a Convolution? — 3Blue1Brown

<https://www.youtube.com/watch?v=KuXjwB4LzSA>

★ Priority · Phase 1. Animated visual intuition for the convolution operation, connecting probability, image processing, and signal processing views of the same operation. ~23 min.

► Attention Is All You Need (Paper Explained) — Yannic Kilcher

<https://www.youtube.com/watch?v=iDulhoQ2pro>

Phase 5. Deep walkthrough of the original Transformer paper: multi-head attention, positional encoding, and why the architecture generalises so well. Prerequisite intuition for Vision Transformers.

► An Image Is Worth 16×16 Words — Yannic Kilcher

https://www.youtube.com/watch?v=TrdevFK_am4

★ Priority · Phase 5. Full walkthrough of the Vision Transformer paper: patch embedding, the [CLS] token, and why ViT overtakes CNNs at scale.

► OpenAI CLIP: Connecting Text and Images (Paper Explained) — Yannic Kilcher

<https://www.youtube.com/watch?v=T9XSU0pKX2E>

★ Priority · Phase 5. Covers contrastive pretraining, the shared embedding space, and how zero-shot classification falls out of a similarity computation between text and image features.

Online Courses

■ CS231n: Convolutional Neural Networks for Visual Recognition — Stanford

<https://cs231n.github.io/convolutional-networks/>

★ Priority · Phase 1–3. Free lecture notes from Stanford's computer vision course (originally taught by Andrej Karpathy). The CNN, detection, and segmentation notes are some of the clearest technical writing available on the topic.

■ Kaggle Learn: Computer Vision

<https://www.kaggle.com/learn/computer-vision>

Phase 1. Short interactive course covering CNN building blocks, transfer learning, and data augmentation, with executable notebooks. Good hands-on complement to CS231n.

Textbooks & Long-Form Reading

● Dive into Deep Learning (d2l.ai) — Zhang, Lipton, Li & Smola

<https://d2l.ai/>

★ Priority · All phases. Free interactive textbook with runnable PyTorch code. Key chapters: [Ch. 7 \(CNNs\)](#), [Ch. 14 \(Object Detection & Segmentation\)](#), and [Ch. 11 \(Attention & Transformers\)](#).

● Deep Learning — Goodfellow, Bengio & Courville

<https://www.deeplearningbook.org/>

Phase 1. Chapter 9 (Convolutional Networks) is the standard theoretical reference for why convolution, pooling, and weight sharing work. Free online.

● Understanding Deep Learning — Simon J.D. Prince

<https://udlbook.github.io/udlbook/>

Phase 5. Chapter 10 (Transformers) is one of the most accessible derivations of self-attention and the Vision Transformer available. Free PDF.

Seminal Papers

■ Deep Residual Learning for Image Recognition (ResNet) — He et al., 2015

<https://arxiv.org/abs/1512.03385>

Phase 1. Introduces skip connections, enabling networks with 50+ layers to train reliably. The backbone architecture used throughout this module.

■ An Image Is Worth 16×16 Words (ViT) — Dosovitskiy et al., 2020

<https://arxiv.org/abs/2010.11929>

Phase 5. The original Vision Transformer paper — patch embedding and pure-attention image classification at scale.

■ Learning Transferable Visual Models From Natural Language Supervision (CLIP) — Radford et al., 2021

<https://arxiv.org/abs/2103.00020>

Phase 5. Introduces contrastive image-text pretraining and zero-shot transfer.

■ Grad-CAM: Visual Explanations from Deep Networks — Selvaraju et al., 2019

<https://arxiv.org/abs/1610.02391>

Phase 1. The gradient-weighted class activation mapping technique used for explainability throughout this module.

■ Segment Anything (SAM) — Kirillov et al., 2023

<https://arxiv.org/abs/2304.02643>

Phase 3. Foundation model for promptable image segmentation — the current state of the art in general-purpose segmentation.

Kaggle Tutorials — Hands-On Practice

◆ Kaggle Learn: Computer Vision

<https://www.kaggle.com/learn/computer-vision>

★ Priority · Phase 1. Interactive lessons on CNN building blocks, transfer learning, and data augmentation with executable notebooks.

◆ CIFAR-10 — Object Recognition in Images

<https://www.kaggle.com/c/cifar-10>

Phase 1. The exact dataset used throughout this module, as a Kaggle competition with a public leaderboard — useful for benchmarking your own model.

◆ Dogs vs. Cats

<https://www.kaggle.com/c/dogs-vs-cats>

Phase 1. Classic binary transfer-learning benchmark. Good for practicing the freeze-backbone / replace-head workflow on a different dataset.

Kaggle Competition Datasets — For Extension Projects

Extension dataset guidance: The module uses CIFAR-10 and COCO-pretrained models throughout. These Kaggle datasets are for independent projects and further practice. Each tests a different aspect of the CV stack (detection, segmentation, small-object detection). A Kaggle account is required.

■ Global Wheat Detection

<https://www.kaggle.com/c/global-wheat-detection>

Extension. Bounding-box detection of wheat heads in outdoor field images. Tests: small, dense, overlapping object detection — a harder case than COCO.

■ Carvana Image Masking Challenge

<https://www.kaggle.com/c/carvana-image-masking-challenge>

Extension. Pixel-precise car segmentation masks. Tests: high-precision binary segmentation — the U-Net architecture was popularised here.

■ Airbus Ship Detection Challenge

<https://www.kaggle.com/c/airbus-ship-detection>

Extension. Satellite imagery ship segmentation with heavy class imbalance (most images contain no ships). Tests: segmentation under extreme imbalance.

Official Documentation

§ torchvision.models

<https://pytorch.org/vision/stable/models.html>

Phase 1 — Essential. Reference for all pretrained model architectures (ResNet, Faster R-CNN, DeepLabv3+, ViT) — weight names and required transforms.

§ timm (PyTorch Image Models)

<https://huggingface.co/docs/timm/index>

Phase 1. Hugging Face library with 400+ pretrained vision models. Use `timm.list_models('vit_*')` to browse available ViT variants.

§ pytorch-grad-cam

<https://github.com/jacobgil/pytorch-grad-cam>

Phase 1. Production-ready GradCAM, GradCAM++, and EigenCAM implementations for reference beyond the from-scratch version built in this module.

§ Ultralytics YOLOv8 Documentation

<https://docs.ultralytics.com/>

Phase 2. Modern single-stage detector docs. Useful contrast with Faster R-CNN for the two-stage vs single-stage detector discussion.

§ OpenAI CLIP

<https://github.com/openai/CLIP>

Phase 5. Model card, code, and usage examples for the CLIP model family.

§ ONNX Runtime

<https://onnxruntime.ai/docs/>

Phase 5. Inference runtime for ONNX models — API docs for InferenceSession and SessionOptions across CPU, GPU, and mobile targets.

§ PyTorch: Exporting a Model to ONNX

https://pytorch.org/tutorials/beginner/onnx/export_simple_model_to_onnx_tutorial.html

Phase 5. Official tutorial for `torch.onnx.export`, dynamic axes, and verifying an exported model with onnxruntime.

AI PROMPTS & CRITIQUE GUIDE

How to use these prompts: Paste each prompt into [Claude](#) or [ChatGPT](#). Apply the **Critique / Verify** guidance. If the model makes an error, force it to fix it. Write a short note about what each model got right and wrong — this is itself a valuable learning exercise.

Phase 1 — GradCAM Deep-Dive

Ask: "Explain, step by step, how GradCAM produces a class activation map from a convolutional neural network. Use concrete tensor shapes (e.g., a ResNet-50 layer4 output of shape [1, 2048, 7, 7]) and show the equations. Then explain why ReLU is applied at the end."

Critique / Verify: Must correctly state that gradients are averaged over spatial dims (H, W) to get per-channel weights: `weights = grad.mean([2,3])`, `cam = relu(sum(weights * acts, dim=1))`. Must explain the ReLU as discarding negative evidence (channels that suppress the target class). Push back if it conflates GradCAM with vanilla CAM, which requires global average pooling baked into the original architecture.

Phase 1 — Transfer Learning Theory

Ask: "Why does freezing the backbone and only training the final classification head work well for transfer learning from ImageNet to a new dataset? What do the early vs late layers of a CNN learn, and which layers would you expect to be most task-specific?"

Critique / Verify: Must state early layers learn edges/textures (task-agnostic) while later layers learn task-specific semantics. Must identify the last conv block as most task-specific. Must acknowledge the domain gap between pretraining and target data and why transfer still works despite it. Push back on any claim that you should always fine-tune the entire model — freezing is optimal when data is small.

Phase 4 — VAE Reparameterisation Trick

Ask: "Explain the reparameterisation trick in variational autoencoders. Why can we not simply sample $z \sim N(\mu, \sigma^2)$ directly in the forward pass and backpropagate through it? How does the trick fix this?"

Critique / Verify: Must state that sampling is non-differentiable — gradients cannot flow through a stochastic node. Must show $z = \mu + \sigma * \epsilon$, $\epsilon \sim N(0,1)$, making ϵ the random variable and z a deterministic function of the learned parameters. Push back if it conflates the reparameterisation trick with the ELBO itself — related but distinct concepts.

Phase 5 — ViT vs. CNN Inductive Biases

Ask: "Compare the inductive biases built into convolutional neural networks vs. Vision Transformers. Under what conditions does ViT outperform CNN-based models, and under what conditions does a CNN still win?"

Critique / Verify: Must list CNNs' inductive biases: translation equivariance and locality. Must note ViT has no built-in spatial bias beyond positional encoding — it learns spatial structure from data. Must correctly state ViT wins primarily at large scale (large datasets and models); CNNs still win on small data. Push back on any claim that ViT is unconditionally superior — this is scale-dependent.

MID-WEEK DISCUSSION QUESTIONS

How to use: Write a 2–3 sentence answer before the session. During discussion, compare reasoning with peers. The quality of your justification matters more than the conclusion.

Q1. Explainability vs Accuracy

[Phase 1] GradCAM shows a classifier attending to background pixels rather than the object itself, yet its held-out accuracy is high. Is this a data problem, a model problem, or both? What intervention would most effectively fix it, and would that intervention risk lowering accuracy?

Q2. Zero-Shot vs Fine-Tuned

[Phase 1 & 5] You have 500 new images of a product category with no labels. Would you reach for a fine-tuned CNN classifier or a zero-shot vision-language model first? What accuracy threshold or business constraint would change your answer?

Q3. NMS Threshold Trade-offs

[Phase 2] Non-Maximum Suppression removes overlapping boxes by an IoU threshold. What is the failure mode when the threshold is set too high? Too low? How would you tune it for a scene with many small, closely packed objects?

Q4. Classical vs Learned Segmentation

[Phase 3] You have a strict per-frame latency budget, controlled lighting, and a single well-defined object type. When does a classical (non-learned) segmentation pipeline outperform a deep segmentation model in practice? What changes your answer?

Q5. GAN vs VAE for Synthetic Data

[Phase 4] A colleague proposes a GAN instead of a VAE for generating synthetic training images. What are the concrete training-stability trade-offs? For a small, low-data class, which would you recommend and why?

Q6. When Zero-Shot Wins

[Phase 5] A zero-shot vision-language model achieves noticeably lower accuracy than a fully supervised model trained on thousands of labelled examples. Is there a realistic scenario where the lower-accuracy zero-shot model is still the better production choice?

Q7. From Notebook to Production

[Phase 5] You have exported a trained classifier to ONNX. A product manager asks whether it can be deployed for real-time inference in a browser or on a mobile device. What technical steps are required, and what accuracy or latency risks should you flag?

PHASE → RESOURCE MAP

Phase	Topic	★ Watch / Practice	★ Read
1	CNN Foundations & Classification	Karpathy — Backprop from scratch 3B1B — What is a convolution? Kaggle Learn: CV	CS231n: ConvNets d2l.ai Ch. 7 ResNet paper
2	Object Detection	Global Wheat Detection	d2l.ai Ch. 14 Ultralytics YOLOv8 Docs
3	Image Segmentation	Carvana Masking Airbus Ship Detection	d2l.ai Ch. 14 Segment Anything (SAM) paper

Phase	Topic	★ Watch / Practice	★ Read
4	Generative Models — VAE	—	Goodfellow Ch. 20 (Generative Models)
5	Vision Transformers & Deployment	Kilcher — ViT paper Kilcher — CLIP paper	Prince Ch. 10 (Transformers) ViT paper CLIP paper